

Table of No-show Indication by Columns

Patient Analysis (pdf) - columns vs No-show

No.	Column Names	Indication
1	age_no_show	Very weak
2	gender_no_show	Very weak
3	city_no_show	Very weak
4	neighborhood_no_show	Very weak
5	appointment_day	Moderate
6	waiting_days	Very Strong
7	appointment_hour	Moderate
8	Employment_status	Very Weak
9	Education_level	Very Weak
10	Income_level	Strong
11	children_count	Very Weak
12	patient_satisfaction_score	Very Strong

Patient Analysis 2 (pdf2) - columns vs No_show

No.	Column Names	Indication
13	Day-of-week	Very Weak
14	Months	Weak
15	sms_message	Strong
16	reminder calls	very strong
17	previous appointments	Very Weak
18	previous_no_shows	Very Strong

Clinic Analysis (cdf2) - columns vs No_show

No.	Column Names	Indication
1	Distance Km	Moderate to Strong
2	Transportation type	Moderate
3	Insurancce_type	Strong
4	clinic_type	Weak
5	weather_condition	Strong
6	temperature_celsius	Very Weak

Medical Analysis (cdf2) - columns vs No_show

No.	Column Names	Indication
1	medical_conditions	Weak
2	appointment_type	Weak
3	doctor_specialty	Very Weak

Data Exploration and Transformation

1. Age values equal to 0 were treated as invalid because records existed with age = 0 and adult characteristics (e.g., divorced marital status). These values were converted to missing and imputed using the median age.
2. Age values which are less than 18 having adult characteristics (e.g., divorced marital status). These Values contain inconsistencies. Data quality checks revealed inconsistent records, such as patients aged 9 marked as retired or married. These records were treated as data quality issues
3. Data quality assessment identified unrealistic combinations (e.g., patients under 18 marked as married, retired, or having children), suggesting that some features may be synthetically generated or contain inconsistencies
4. Missing values OF 1500 in categorical variables such as education_level were imputed using the category "Unknown" to preserve records while explicitly indicating unavailable information.

Difference in No-show Rate	Typical Interpretation
< 1%	Very Weak
1% - 2%	Weak
2% - 3%	Moderate
3% - 5%	Strong
> 5%	Very Strong

Patient Analysis (pdf) - columns vs No_show

1. age_no_show (Very weak): Age-group analysis revealed only minor differences in no-show rates across age categories. Patients aged 65-100 recorded the highest no-show rate (8.52%), followed by those aged 0-18 (8.41%) and 50-65 (8.38%). Patients aged 35-50 exhibited the lowest no-show rate (8.04%). Overall, no-show rates ranged from 8.04% to 8.52%, suggesting that age group has a limited influence on appointment attendance.

no_show	0	1	0_pct	1_pct
age_group				
0-18	5307	487	91.59%	8.41%
18-35	11245	1016	91.71%	8.29%
35-50	13812	1207	91.96%	8.04%
50-65	9214	843	91.62%	8.38%
65-100	6284	585	91.48%	8.52%

2. gender_no_show (Very weak): Gender appears to have little influence on appointment attendance. Female patients recorded a no-show rate of 8.43%, compared to 8.08% for males and 8.74% for the 'Other' category. The differences between groups are small, suggesting that gender is not a strong predictor of no-show behaviour.

no_show	0	1
gender		
Female	91.57%	8.43%
Male	91.92%	8.08%
Other	91.26%	8.74%

3. **city_no_show** (Very weak): City-wise analysis shows only minor variation in no-show rates across locations. Tilburg recorded the highest no-show rate at 8.82%, followed by Groningen (8.57%) and Almere (8.54%). Rotterdam had the lowest no show rate at 8.01%, closely followed by Amsterdam (8.02%). Overall, no-show rates remained within a narrow range of 8.01% to 8.82%, suggesting that city has a limited influence on appointment attendance.

no_show	0	1	0_pct	1_pct
city				
Almere	4154	388	91.46%	8.54%
Amsterdam	8223	717	91.98%	8.02%
Eindhoven	5598	494	91.89%	8.11%
Groningen	4116	386	91.43%	8.57%
Rotterdam	7361	641	91.99%	8.01%
The Hague	6392	589	91.56%	8.44%
Tilburg	4126	399	91.18%	8.82%
Utrecht	5892	524	91.83%	8.17%

4. **neighborhood_no_show** (Very weak): Neighborhood-wise analysis reveals modest variation in no-show rates across regions. Suburban areas recorded the highest no-show rate at 8.75%, followed by East (8.41%), Central (8.38%), and Rural (8.38%). North exhibited the lowest no-show rate at 7.77%, with West also showing a relatively low rate of 7.91%. Overall, no-show rates ranged from 7.77% to 8.75%, indicating that neighborhood may have a slight influence on appointment attendance, although the differences are not substantial.

no_show	0	1	0_pct	1_pct
neighborhood				
Central	8256	755	91.62%	8.38%
East	6351	583	91.59%	8.41%
North	5995	505	92.23%	7.77%
Rural	4593	420	91.62%	8.38%
South	5932	526	91.86%	8.14%
Suburban	8286	795	91.25%	8.75%
West	6449	554	92.09%	7.91%

5. **appointment_day (Moderate):** The no-show rate varied across days of the month, ranging from 6.90% to 9.42%. Day 18 recorded the highest no-show rate (9.42%), followed by days 1 (9.38%) and 12 (9.21%). In contrast, day 21 had the lowest no-show rate (6.90%), followed by days 6 (6.97%) and 17 (7.08%). The variation of approximately 2.5 percentage points suggests that the day of the month may have a moderate influence on appointment attendance.

Top 5 appointment_day

no_show_over_year	show	no_show	total	1_pct
day				
18	1462	152	1614	9.42
1	1516	157	1673	9.38
12	1538	156	1694	9.21
7	1502	152	1654	9.19
5	1486	150	1636	9.17

Least 5 appointment_day

24	1518	123	1641	7.50
28	1553	124	1677	7.39
17	1497	114	1611	7.08
6	1481	111	1592	6.97
21	1539	114	1653	6.90

6. **waiting_days (Very strong):** Patients with longer waiting periods tend to have higher no-show rates. The no-show rate rises from 5.9% for appointments scheduled within 0-7 days to 15.3% for appointments scheduled 31-60 days in advance. The patients waiting more than 60 days exhibited the highest no-show rate (30.00%), although this group contained a relatively smaller number of appointments (30 total records).

no_show	0	1
waiting_group		
0-7	94.09%	5.91%
8-14	92.11%	7.89%
15-21	90.52%	9.48%
22-30	88.39%	11.61%
31-60	84.68%	15.32%
61+	70.0%	30.0%

7. **appointment_hour (Moderate):** Appointment attendance varied by time slot. Evening appointments (5 PM to 9 PM) recorded the highest no-show rate at 10.22%, compared with 8.23% for morning appointments and 8.15% for afternoon appointments. Although evening appointments represented a smaller share of total bookings, they exhibited a noticeably higher likelihood of no-shows.

no_show	0	1	0_pct	1_pct
appointment_hour_group				
6Am-12Am	24348	2183	91.77%	8.23%
12Pm-5Pm	19714	1750	91.85%	8.15%
5pm-9Pm	1800	205	89.78%	10.22%

8. **Employment_status (Very weak):** Employment status shows only minor differences in appointment attendance. No-show rates were relatively consistent across employment groups, Unemployed patients recorded the highest no-show rate at 8.84%, followed by employed patients at 8.28%. Students had the lowest no-show rate at 7.89%, while retired (8.20%) and self-employed (8.09%) patients exhibited similar attendance patterns. Overall, no-show rates ranged from 7.89% to 8.84%, suggesting that employment status has a limited influence on no-show behaviour.

	no_show	0	1	0_pct	1_pct
employment_status					
Unemployed	6288	610	91.16%	8.84%	
Employed	21248	1918	91.72%	8.28%	
Retired	8207	733	91.8%	8.2%	
Self-employed	4524	398	91.91%	8.09%	
Student	5595	479	92.11%	7.89%	

9. **Education_level (Very weak):** Education level exhibits some variation in no-show behaviour. Patients with an unknown education level recorded the highest no-show rate (8.67%), followed by Bachelor's degree holders (8.47%) and those with secondary education (8.45%). Master's degree holders had the lowest no-show rate (7.81%), while patients with primary, vocational, and PhD-level education showed similar attendance patterns. Overall, no-show rates ranged from 7.81% to 8.67%, suggesting that education level may have a modest influence on appointment attendance.

	no_show	0	1	0_pct	1_pct
education_level					
Unknown	1370	130	91.33%	8.67%	
Bachelor	10135	938	91.53%	8.47%	
Secondary	12291	1134	91.55%	8.45%	
Vocational	10720	954	91.83%	8.17%	
PhD	937	83	91.86%	8.14%	
Primary	5441	478	91.92%	8.08%	
Master	4968	421	92.19%	7.81%	

10. **Income_level (Strong):** Income level demonstrates a noticeable association with appointment attendance. Patients in the Low-income group recorded the highest no-show rate at 10.65%, substantially higher than all other income categories. In contrast, Middle-income (7.17%) and Upper-Middle-income (7.23%) patients exhibited the lowest no-show rates. High-income patients also showed relatively low no-show rates (7.55%). Overall, the results suggest that lower income levels are associated with a greater likelihood of missing appointments.

no_show	0	1	0_pct	1_pct
income_level				
Low	7756	924	89.35%	10.65%
Lower-Middle	10549	1034	91.07%	8.93%
Unknown	1824	176	91.2%	8.8%
High	2596	212	92.45%	7.55%
Upper-Middle	8038	626	92.77%	7.23%
Middle	15099	1166	92.83%	7.17%

11. **children_count** (Very weak): The number of children a patient has appears to have a limited impact on appointment attendance. Patients with two children recorded the highest no-show rate (8.55%), while patients with four children had the lowest rate (8.10%). Across all groups, no-show rates remained within a narrow range of 8.10% to 8.55%, suggesting that the number of children is not a strong predictor of no-show behaviour.

no_show	0	1	0_pct	1_pct
children_count				
2	9970	932	91.45%	8.55%
5	884	80	91.7%	8.3%
1	9830	884	91.75%	8.25%
3	5467	491	91.76%	8.24%
0	17420	1549	91.83%	8.17%
4	2291	202	91.9%	8.1%

12. **patient_satisfaction_score** (Very strong): Patient satisfaction demonstrated a strong inverse relationship with no-show behaviour. The no-show rate decreased consistently from 21.43% among patients with satisfaction scores between 0 and 1 to 4.93% among those with scores between 9 and 10. This trend suggests that higher patient satisfaction is strongly associated with improved appointment attendance. The 0-1 satisfaction group contained relatively few observations and should be interpreted with caution; however, the overall downward trend remains consistent across the remaining score bands.

no_show	0	1	0_pct	1_pct
patient_satisfaction_group				
0-1	11	3	78.57%	21.43%
1-2	76	12	86.36%	13.64%
2-3	342	52	86.8%	13.2%
3-4	1239	199	86.16%	13.84%
4-5	3166	430	88.04%	11.96%
5-6	6520	751	89.67%	10.33%
6-7	9186	876	91.29%	8.71%
7-8	10607	909	92.11%	7.89%
8-9	7564	535	93.39%	6.61%
9-10	7151	371	95.07%	4.93%

Patient Analysis 2 (pdf2) - columns vs No_show

13. Day-of-week (Very weak): Day-of-week analysis revealed only minor variation in no-show rates. Thursday recorded the highest no-show rate (8.62%), followed by Monday (8.55%), while Saturday had the lowest no-show rate (7.90%). Overall, no-show rates ranged from 7.90% to 8.62%, suggesting that the day of the week has a limited influence on appointment attendance.

no_show	0	1	0_pct	1_pct
day_of_week				
Monday	6539	611	91.45	8.55
Tuesday	6691	594	91.85	8.15
Wednesday	6476	591	91.64	8.36
Thursday	6470	610	91.38	8.62
Friday	6577	577	91.93	8.07
Saturday	6482	556	92.10	7.90
Sunday	6627	599	91.71	8.29

14. Months (Weak): Monthly analysis revealed moderate variation in no-show rates throughout the year. June recorded the highest no-show rate (8.95%), followed by February (8.94%) and August (8.70%), while July had the lowest no-show rate (7.70%). Overall, no-show rates ranged from 7.70% to 8.95%, indicating that month of appointment may have a modest influence on attendance behaviour.

no_show	0	1	0_pct	1_pct
Jan	3855	358	91.50	8.50
Feb	3512	345	91.06	8.94
Mar	3894	358	91.58	8.42
Apr	3688	322	91.97	8.03
May	4003	356	91.83	8.17
Jun	3694	363	91.05	8.95
Jul	3874	323	92.30	7.70
Aug	3842	366	91.30	8.70
Sep	3795	324	92.13	7.87
Oct	3958	364	91.58	8.42
Nov	3806	327	92.09	7.91
Dec	3941	332	92.23	7.77

15. sms_message: (Strong): Patients who did not receive an SMS reminder exhibited a substantially higher no-show rate (10.94%) compared to those who received an SMS reminder (7.25%). This suggests that appointment reminders may play an important role in reducing missed appointments.

no_show	0	1	0_pct	1_pct
sms_received				
not_recived	12393	1522	89.06	10.94
recived	33469	2616	92.75	7.25

16. reminder calls: (Very strong): A strong negative relationship was observed between the number of reminder calls and no-show rates. Patients who received no reminder calls had a no-show rate of 9.66%, which steadily decreased to 8.15%, 6.47%, and 5.17% for patients receiving one, two, and three reminder calls, respectively. This suggests that reminder calls may be highly effective in improving appointment attendance.

no_show	0	1	0_pct	1_pct
reminder_calls				
0	17314	1851	90.34%	9.66%
1	17780	1577	91.85%	8.15%
2	8366	579	93.53%	6.47%
3	2402	131	94.83%	5.17%

17. previous appointments (Very weak) The number of previous appointments showed minimal variation in no-show rates. Patients with 0-4 previous appointments recorded a no-show rate of 8.21%, compared to 8.41% among those with 8-16 previous appointments. The narrow range of no-show rates suggests that appointment history has little influence on attendance behaviour.

no_show	0	1	0_pct	1_pct
previous_appointments_group				
0-4	20316	1817	91.79%	8.21%
4-8	22453	2037	91.68%	8.32%
8-16	3093	284	91.59%	8.41%

18. Previous_no_shows: (Very strong): Strong positive relationship was observed between previous no-show history and future no-show behavior. Patients with no prior no-shows recorded a no-show rate of 5.92%, which increased steadily as the number of previous missed appointments rose, reaching 26.53% among patients with seven prior no-shows and 31.82% among those with eight or more prior no-shows. This suggests that past no-show behavior is a strong predictor of future appointment attendance.

	no_show	0	1	total	0_pct	1_pct
previous_no_shows						
	0	24315	1530	25845	94.08%	5.92%
	1	12414	1160	13574	91.45%	8.55%
	2	5501	749	6250	88.02%	11.98%
	3	2326	412	2738	84.95%	15.05%
	4	837	172	1009	82.95%	17.05%
	5	317	70	387	81.91%	18.09%
	6	101	25	126	80.16%	19.84%
	7	36	13	49	73.47%	26.53%
	8+	15	7	22	68.18%	31.82%

Clinic Analysis (cdf2) - columns vs No_show

1. Distance Km (Moderate to Strong): Distance to the healthcare facility showed a positive relationship with no-show behaviour. Patients living within 2 km recorded the lowest no-show rate (6.86%), while those traveling more than 20 km had the highest rate (10.93%). Overall, no-show rates increased steadily as travel distance increased, suggesting that longer travel distances may contribute to missed appointments.

	no_show	0	1	0_pct	1_pct
distance_km_group					
0-2km	4249	313	93.14%	6.86%	
2-5km	12351	976	92.68%	7.32%	
5-10km	16441	1468	91.8%	8.2%	
10-20km	11020	1160	90.48%	9.52%	
20-65+km	1801	221	89.07%	10.93%	

2. Transportation type (Moderate): Transportation type showed moderate variation in no-show rates. Patients who travelled by walking recorded the highest no-show rate (9.68%), followed by those with unknown transportation methods (9.20%) and public transport users (9.00%). In contrast, patients traveling by car exhibited the lowest no-show rate (7.27%), followed by taxi users (7.76%). These findings suggest that transportation mode may have some influence on appointment attendance.

	no_show	0	1	0_pct	1_pct
transportation_type					
Bike	7279	651	91.79%	8.21%	
Car	16031	1257	92.73%	7.27%	
Public Transport	14352	1419	91.0%	9.0%	
Taxi	2663	224	92.24%	7.76%	
Unknown	1135	115	90.8%	9.2%	
Walking	4402	472	90.32%	9.68%	

3. **Insurance_type (Strong):** Insurance type exhibited a noticeable association with appointment attendance. Uninsured patients recorded the highest no-show rate (11.04%), substantially higher than all other insurance categories. In contrast, patients with private insurance had the lowest no-show rate (7.60%), followed closely by those with mixed insurance coverage (7.67%). These findings suggest that insurance status may be an important factor influencing no-show behaviour.

	no_show	0	1	0_pct	1_pct
insurance_type					
Mixed	4476	372	92.33%	7.67%	
Private	11815	972	92.4%	7.6%	
Public	23640	2077	91.92%	8.08%	
Uninsured	5247	651	88.96%	11.04%	
Unknown	684	66	91.2%	8.8%	

4. **clinic_type (Weak):** No-show rates varied modestly across clinic types, ranging from 7.25% at private hospitals to 8.60% at community clinics. Public hospitals (8.59%) and community clinics (8.60%) exhibited the highest no-show rates, while private hospitals recorded the lowest. The relatively small differences suggest that clinic type has a limited to moderate influence on appointment attendance.

	no_show	0	1	0_pct	1_pct
clinic_type					
Community Clinic	11428	1075	91.4%	8.6%	
Private Hospital	8450	661	92.75%	7.25%	
Public Hospital	19163	1800	91.41%	8.59%	
Specialty Clinic	6821	602	91.89%	8.11%	

5. **weather_condition (Strong):** Weather conditions showed a strong association with appointment attendance. Patients scheduled during rainy conditions recorded the highest no-show rate (10.43%), followed by stormy (9.05%) and snowy weather (8.65%). In contrast, sunny (7.48%) and cloudy conditions (7.31%) exhibited the lowest no-show rates. Overall, no-show rates increased as weather conditions became more severe, suggesting that adverse weather may contribute to missed appointments.

	no_show	0	1	0_pct	1_pct
weather_condition					
Cloudy	14797	1167	92.69%	7.31%	
Rainy	10743	1251	89.57%	10.43%	
Snowy	1795	170	91.35%	8.65%	
Stormy	2774	276	90.95%	9.05%	
Sunny	15753	1274	92.52%	7.48%	

6. **temperature_celsius (Very weak):** Temperature showed little association with no-show behaviour. No-show rates remained relatively stable across temperature ranges, varying from 7.71% to 8.42%. The absence of a clear trend suggests that temperature is not a strong predictor of appointment attendance.

	no_show	0	1	0_pct	1_pct
temperature_celsius_group					
-10-0 c	1843	154	92.29%	7.71%	
0-10 c	12349	1136	91.58%	8.42%	
10-20 c	21379	1915	91.78%	8.22%	
20-30 c	9241	840	91.67%	8.33%	
30-40 c	1050	93	91.86%	8.14%	

Medical Analysis (cdf2) - columns vs No_show

1. **medical_conditions (Weak):** Medical conditions exhibited only a weak association with appointment attendance. Patients with disabilities recorded the highest no-show rate (9.17%), followed by those with alcoholism (8.61%) and hypertension (8.50%). Patients with diabetes (8.02%) and chronic diseases (7.78%) showed slightly lower no-show rates. Overall, the variation across medical conditions was relatively small, with a difference of only 1.39 percentage points between the highest and lowest groups, suggesting that medical conditions are not major drivers of no-show behaviour.

	0_pct	1_pct
has_chronic_disease	91.42	7.78
hypertension	91.83	8.50
diabetes	91.65	8.02
alcoholism	91.76	8.61
disability	91.79	9.17

	has_chronic_disease	hypertension	diabetes	alcoholism	disability
has_chronic_disease	1.000000	0.182680	0.161974	0.001049	-0.002965
hypertension	0.182680	1.000000	0.088182	0.007126	-0.003510
diabetes	0.161974	0.088182	1.000000	0.002239	0.002373
alcoholism	0.001049	0.007126	0.002239	1.000000	0.001646
disability	-0.002965	-0.003510	0.002373	0.001646	1.000000

2. **appointment_type (Weak):** Appointment type demonstrated a strong relationship with no-show behavior. Emergency appointments recorded the lowest no-show rate (4.87%), substantially lower than all other appointment categories. In contrast, therapy appointments exhibited the highest no-show rate (9.02%), followed by diagnostic tests (8.90%) and routine checkups (8.78%). The 4.15 percentage point difference between the highest and lowest categories suggests that appointment type is an important factor influencing patient attendance.

The pattern also makes business sense in ways as:

- Emergency appointments have the lowest no-show rate because patients usually need immediate care.
- Follow-up appointments are also relatively important, leading to better attendance.
- Therapy, Diagnostic Tests, and Routine Checkups are more likely to be postponed or skipped.

no_show	0	1	0_pct	1_pct
appointment_type				
Consultation	13726	1315	91.26%	8.74%
Diagnostic Test	6452	630	91.1%	8.9%
Emergency	2850	146	95.13%	4.87%
Follow-up	11061	903	92.45%	7.55%
Routine Checkup	8123	782	91.22%	8.78%
Therapy	3650	362	90.98%	9.02%

3. **doctor_specialty** (Very weak): Doctor specialty showed a limited association with appointment attendance. No-show rates ranged from 7.63% in Endocrinology to 8.70% in General Practice. Cardiology (8.65%) also exhibited relatively high no-show rates, while Neurology (7.65%), Pediatrics (7.86%), and Dermatology (7.90%) recorded lower rates. Overall, the variation across specialties was modest, suggesting that doctor specialty is not a major factor influencing no-show behaviour.

no_show	0	1	0_pct	1_pct
doctor_specialty				
Cardiology	4500	426	91.35%	8.65%
Dermatology	4711	404	92.1%	7.9%
Endocrinology	3258	269	92.37%	7.63%
General Practice	14545	1386	91.3%	8.7%
Neurology	3246	269	92.35%	7.65%
Ophthalmology	3148	289	91.59%	8.41%
Orthopedics	4662	428	91.59%	8.41%
Pediatrics	4058	346	92.14%	7.86%
Psychiatry	3734	321	92.08%	7.92%